

Complementary regional energy features for spoofed speech detection

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ABSTRACT

Automatic speaker verification systems are found to be vulnerable to spoof attacks such as voice conversion, text-to-speech, and replayed speech. As the security of biometric systems is vital, many countermeasures have been developed for spoofed speech detection. To satisfy the recent developments on speech synthesis, publicly available datasets became more and more challenging (e.g., ASVspoof 2019 and 2021 datasets). A variety of replay attack configurations were also considered in those datasets, as they do not require expertise, hence easily performed. This work utilizes regional energy features, which are experimentally proven to be more effective than the traditional frame-based energy features. The proposed energy features are independent from the utterance length and are extracted over nonoverlapping time-frequency regions of the magnitude spectrum. Different configurations are considered in the experiments to verify the regional energy features' contribution to the performance. First, light convolutional neural network – long short-term memory (LCNN – LSTM) model with linear frequency cepstral coefficients is used to determine the optimal number of regional energy features. Then, SE-Res2Net model with log power spectrogram features is used, which achieved comparable results to the state-of-the-art for ASVspoof 2019 logical access condition. Physical access condition from ASVspoof 2019 dataset, logical access and deep fake conditions from ASVspoof 2021 dataset are also used in the experiments. The regional energy features achieved improvements for all conditions with almost no additional computational or memory loads (less than 1% increase in the model size for SE-Res2Net). The main advantages of the regional energy features can be summarized as i) capturing nonspeech segments, ii) extracting band-limited information. Both aspects are found to be discriminative for spoofed speech detection.

1. Introduction

Automatic speaker recognition systems aim to identify or authenticate a speaker based on his/her voice, and they can be found in many commercial applications (Jain et al., 2016). Despite their ease of use and commercial success, they are not immune to spoofing attacks (Sizov et al., 2015). Recently, several challenges were organized which increased the awareness of the detrimental effects of spoof attacks. Publicly available databases related to those challenges let the researchers develop new countermeasures (CM) and compare them with previous ones on a common data (Delgado et al., 2018, Todisco et al., 2019, Wu et al., 2017, Wu et al., 2015, Korshunov et al., 2016, Yamagishi et al., 2021). These datasets consist of various spoofing attacks such as speech synthesis and voice conversion (logical access scenario, LA), and replayed speech (physical access scenario, PA). Details of the datasets will be given in the

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following sections.

In general, spoof detection problem is considered as a two-class problem, where one class is genuine speech, and other is spoofed speech. Traditionally, Gaussian mixture models (GMM) have been used to model each class (Kinnunen et al., 2017, Sahidullah et al., 2015, Kamble and Patil, 2021, Kumar et al., 2021). Other than this binary approach, multiclass classification can be used by considering different spoof attack types as different classes. In this case, more complex classifiers such as deep learning architectures are usually preferred (Qian et al., 2016, Dinkel et al., 2018, Dinkel et al., 2017). As a matter of fact, recent state of the art CM relies heavily on deep learning systems. The reported results on ASVspoof 2021 challenge prove this fact. A fusion of light convolutional neural networks (LCNN) (Wu et al., 2020, Lavrentyeva et al., 2017) with mel-spectrogram inputs achieved lowest tandem detection cost function (t-DCF) and lowest equal error rate (EER) in ASVspoof 2021 LA task, and a fusion of ResNet18 (He et al., 2016) delivered one of the best performances for PA task (Tomilov et al., 2021). A combination of ResNet18, RawNet2 (Tak et al., 2021), and LCNN achieved the top performance for deep fake (DF) task, which was introduced with the ASVspoof 2021 challenge (Tomilov et al., 2021). Similar observations can be made for the ASVspoof 2019 challenge. Following the summary provided in (Nautsch et al., 2021), almost every single and primary (fusion of subsystems) system includes neural networks. For the LA task, the top performing single system is based on linear frequency cepstral coefficients (LFCCs) and LCNN (Lavrentyeva et al., 2019).

Although most studies focused on developing CM, few studies analyzed the data itself, and reached interesting conclusions. One of the most recent reports indicated that the length of silence periods in the training and test utterances can affect the spoof detection performance (Müller et al., 2021). Even using a single feature (i.e., duration of the leading silence) achieved 15.1% EER. Furthermore, absence of the leading silence reduced the RawNet2 model's performance from 3.6% to 15.5% EER for LA task of 2019 challenge. A similar observation can be found in (Zhang et al., 2021) for LCNN and SENet. Another study was conducted for the ASVspoof 2017 dataset version 2 (Chettri et al., 2020), and some artifacts such as pattern differences, burst click, silence, etc. were identified. These studies indicate that the silence sections may provide hints for CM, whether due to the artifacts or due to the speech synthesis methods which may not accurately synthesize the *silence*. As a matter of fact, several studies mentioned that voice activity detectors were not used to take advantage of the artifacts, the channel information, etc. possibly found in the nonspeech frames (Font et al., 2017, Singh and Pati, 2019, Zhao et al., 2018).

Besides the nonspeech regions, several studies take advantage of frequency regions. In (Rupesh Kumar and Bharathi, 2022), sub-bands are weighted with Gaussian function. (Ren et al., 2021) proposed recalibration of bandpass filters to extract useful features which are fed to temporal convolutional network. Low frequency bins are normalized in (Yang and Das, 2019). Upper half band found to be more informative in (Lavrentyeva et al., 2017). Similarly, higher frequency bands performed better for replay attack detection (Witkowski et al., 2017). Contrary, in (Zhang et al., 2017), lower half band is used for ASVspoof 2015 dataset. Lower half band is also achieved lower EER than higher band and full band configurations in (Zhang et al., 2021) for ASVspoof 2019 LA set. Fusion of sub-bands are shown to be capable of detecting attacks in the development set of ASVspoof 2019 LA, but they performed poor on the evaluation set (Tak et al., 2020). These studies indicate that frequency regions have an impact on the final score. Nevertheless, the reported results showed that different spoofing attacks may exploit different frequency bands, hence it is not clear how to select a band that will perform sufficiently on each condition.

Motivated from the abovementioned facts, this paper argues that the traditional frame-based energy features can be replaced with more effective band-limited regional energy features. In (Zhang et al., 2021), the authors have provided spectrograms of a few utterances (Figure 1 of Zhang et al. (2021)), and mentioned that silent segments at the beginning and end of bonafide speech contain weak stochastic noise while synthesized speech has no silence, no noise, or abnormal noise. In (Jung et al., 2019), the authors argued that high resolution information is necessary to detect replay attacks, where focus on minute distinctions is required. They supported this idea by comparing the spectrograms of bonafide and spoofed speech, indicating subtle differences in some regions (Figure 1 of Jung et al. (2019)). Also, for the ASVspoof 2017 set, environment noise was found to be a clue for PA attacks (Delgado et al., 2018). These artifacts cannot be captured by frame-based energy features, as they do not discriminate frequency regions and do not have temporal information. Also, considering cepstral features, the frame energy is just a single feature concatenated to a higher dimensional static and/or delta feature set. Therefore, frame-based energy features' influence on the model will be limited. For spectrogram features, the common practice is to use several convolutional layers for their feature extraction capabilities, then apply recurrent and/or dense layers to get class probabilities. This process, however, may also lose useful information since the kernel size of the convolutional layers are usually small (such as 3×3 , 5×5 , 7×7). Also, non-linearity functions such as rectified linear unit (ReLU), max-feature-map (MFM), or max-pooling favor high values. Silent regions will inherently have low values, hence discriminative information may not propagate through deeper layers of the network. In fact, relatively shallow networks are found to be more effective than deeper networks (Hua et al., 2021). Due to these reasons, providing regional energy information to the network will be useful as they are expected to capture complementary discriminative information, especially for synthesized speech.

The proposed energy features are based on 2-dimensional " A_2 " method of (Guido, 2016). Contrary to the conventional frame-based energy features, several adjacent frames and frequency bins are considered based on the selected number of equal rectangular regions. Several previous studies showed that the energy information can be beneficial for spoof detection, such as log energy information with cepstral mean-variance normalization (CMVN) (Delgado et al., 2018), magnitude-phase energy with constant-Q transform (CQT)-based coefficients (Yang et al., 2019), independently modeling of high and low energy frames (Suthokumar et al., 2017), normalizing frame energies of low frequency bins (Yang and Das, 2019), Teager energy operator (Kamble and Patil, 2021, Patil et al., 2022). As the proposed approach localizes on different time-frequency regions, it inherently separates the frequency bands, hence demonstrating another advantage over frame-based energies.

The proposed spoof detection systems use LFCC and log power spectrogram features. LCNN – LSTM [41] and Res2Net with squeeze-excitation (SE) blocks (Li et al., 2021) are used as deep learning models. Since the regional energy features are independent from the

utterance lengths, they are concatenated to the embeddings of the deep models' last layer, hence they are named as complementary features. ASVspoof 2019 and ASVspoof 2021 database are used to examine the performance of the proposed approach. For both LA and PA conditions of ASVspoof 2019, the proposed system achieved the lowest EERs compared to the other LCNN based systems found in the literature. For the LA condition of ASVspoof 2019, SE-Res2Net model with the regional energy features achieved one of the lowest minimum t-DCF values. For LA and DF conditions of ASVspoof 2021, regional energy features provided better scores than the same system without the regional energy features.

2. Extracting regional energy features

Short-term features have been used in many speech-related systems, including spoof detection. Traditionally, the signal energy can be calculated per frame, and the result can be appended to the extracted features for each frame. Dynamic information of the signal energy can be observed with this approach, and seamlessly combined with the features of interest such as mel-frequency cepstral coefficient (MFCC), constant-Q cepstral coefficient (CQCC), etc. On the other hand, for spoof detection, calculating the signal energy on a broader window may be more beneficial to capture spoofing artifacts or channel effects, as discussed in the previous section. Further, instead of considering all frequency bins, focusing on different frequency regions may enhance the anomalies in spoofed speech. These ideas are worth investigating since:

- (i). Many features, or classifiers, benefit from temporal information. Energy features calculated over several frames are expected to be more discriminative than the frame-based ones. The popular CQCC feature is a long-term window transform (Yang et al., 2019) which produces higher frequency resolution at lower frequencies. Context window applied to the frames (Yang and Das, 2020) for adding temporal information into account for feed forward networks. Recurrent neural networks (RNNs), where the past data influence the current state, used in several spoof detection studies (Dua et al., 2021, Gomez-Alanis et al., 2019). CNNs use sliding convolutional filters which captures several frames based on the filter size (Lavrentyeva et al., 2017, Zhang et al., 2017).
- (ii). As some frequency contents are reported to be more important than the others (Witkowski et al., 2017, Zhang et al., 2017), energy concentrated in a frequency region may be more important to detect spoofed speech. For the CNNs fed with spectrums of speech, similar to the previous case, convolutional filters focus on a frequency band based on the filter size.

To extract energy features in a way that focuses on a number of frames and a frequency region, “ A_2 ” method for 2-D signals proposed in (Guido, 2016) is adopted. The main idea of the “ A_2 ” method is to extract energy information in different levels of resolution. Contrary to the ordinary energy features' computation, there is no overlap between windows in this method. Prime numbers are chosen to avoid linear combinations between resolutions, which otherwise provide redundant information. An image texture classification was presented as an application of the “ A_2 ” method, using prime numbers 2, 3, and 5 to construct a 38-dimensional feature vector, and they were used to distinguish barleyrice patterns.

The same method can be used to extract regional energy features to exploit the spoofing artifacts mentioned in the Introduction section. However, the speech spectrogram does not have a specific texture. Using the same 38-dimensional feature set may not suffice for the spoof detection task. The following paragraphs explain why different prime numbers are preferred in this study and give the necessary equations for computation of the regional energy features.

A number of nonoverlapping, equal-sized rectangle windows are used to frame the 2-D signal, which is the magnitude spectrum obtained via short-term Fourier transform (STFT) in this work, then the normalized energies are calculated within each rectangle. Let T be the number of regions for both time and frequency axes. For instance, if $T = 2$, then the frequency is divided into upper and lower

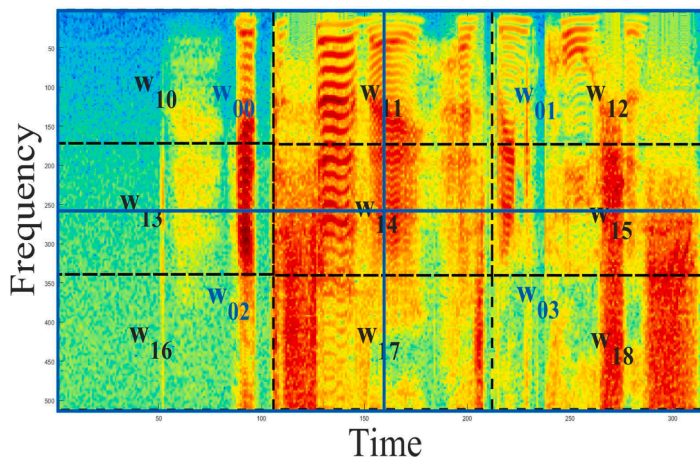


Fig. 1. Placement of equal-sized rectangle windows for $T=2$ (blue) and $T=3$ (black).

half bands. Similarly, the time (frame) axis is divided into two parts. Fig. 1 illustrates this idea with solid blue lines. The upper left rectangle, w_{00} , is the upper half band for the first half of the utterance. w_{01} is the upper half band for the other half of the utterance. w_{02} and w_{03} correspond to the lower half bands. If $T = 3$, then both axes are divided into three segments, illustrated with the black dashed lines on Fig. 1. Now the frequency is divided into three sub-bands, and the utterance is divided into three parts. It should be noted that the number of frequency bins are equal for the whole dataset, hence the frequency resolution will be equal and based on the selected T value. However, the utterance lengths differ, so the time axis of the rectangles are increased or decreased accordingly. This situation results in features independent of the utterance length. The number of features is equal to T^2 . It is also possible to use different values for frequency and time axes, but in this work, they are chosen to be equal.

In (Guido, 2016), “A₂” method is implemented using prime numbers to avoid linear combination of the extracted features, as those features are directly used for image texture classification as an example. However, in this paper, the extracted features will be used as complementary features to the deep embeddings. Therefore, selected T values are {7, 11, 13, 15, 20, 25}. These numbers will create energy feature vectors with lengths {49, 121, 169, 255, 400, 625}, respectively. The reason for choosing these values is to analyze the vector length (and hence frequency resolution) impact on the performance. If T has a small value such as two, the spectrogram will be divided into four regions, which is too broad for speech related tasks as the speech signal is a rapidly varying signal. On the other hand, as T gets bigger, the algorithm will converge to the spectrogram itself. (Refer to Fig. 2 below for a visual representation.) Also, the prime numbers 7, 11, and 13 are chosen to observe the performances of their combinations such as in (Guido, 2016). Considering the smallest chosen value (7) and the biggest chosen value (25), the sub-bands span 73 Hz and 20 Hz per region, respectively for 512 frequency bins with 16 kHz sampling rate. It should be noted that the rectangles may not cover the whole frequency axis or the whole time axis. In that case, a small number of uncovered bins or frames are neglected.

Following the example given in Fig. 1, the normalized energy under the rectangle w_{00} is calculated as

$$\zeta_{00} = \frac{E(w_{00})}{\sum_{k=0}^{T^2-1} E(w_{0k})} \quad (1)$$

where T^2 is equal to 4 for that level ($T=2$), and E is the signal energy defined as

$$E(w) = \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} (w_{i,j})^2 \quad (2)$$

where N is the height, and M is the width of w . So, the regional energy vector of $T=2$ will be

$$\zeta_{T=2} = [\zeta_{00}, \zeta_{01}, \zeta_{02}, \zeta_{03}] \quad (3)$$

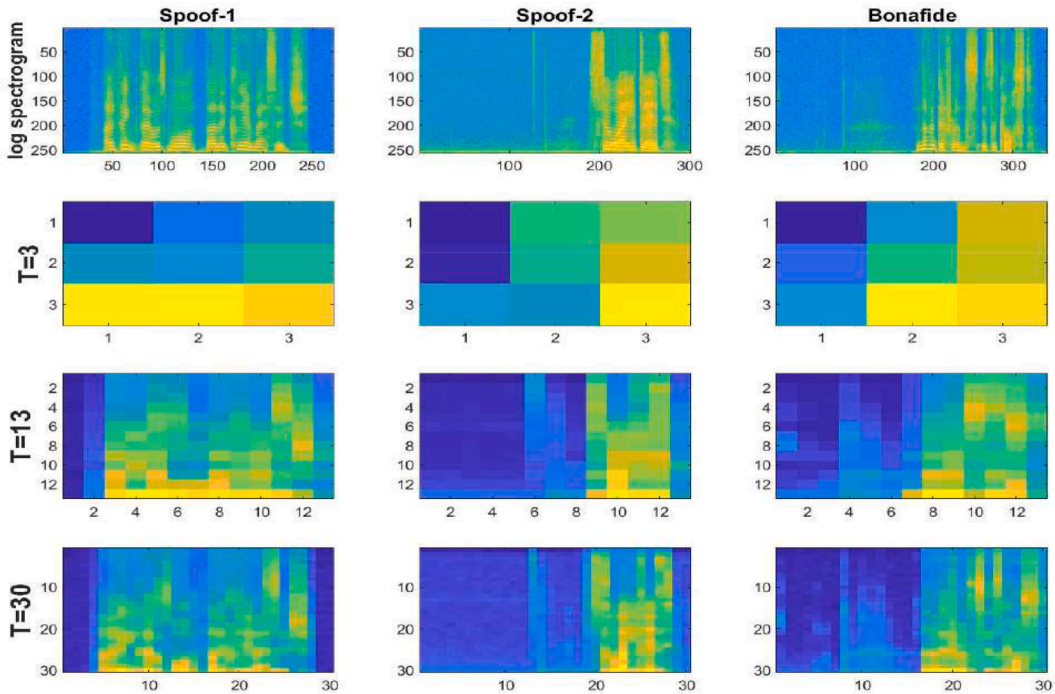


Fig. 2. Spectrograms and regional energy features for randomly chosen spoof and bonafide utterances from ASVspoof 2019 LA training set. For small T values, extracted features are not informative. Higher T values provide detailed information with increased computational and memory loads. Moderate T values may preserve the spoofing artifacts within a compact representation.

The vector for $T=3$ is found as Equation 4, following the same steps.

$$\zeta_{T=3} = [\zeta_{10}, \zeta_{11}, \zeta_{12}, \zeta_{13}, \dots, \zeta_{18}] \quad (4)$$

As T gets bigger, the resulting vector length increases. It is also possible to concatenate vectors extracted with different T values as given below Equation 5. For instance, concatenating vectors of $T=7$ and $T=11$ will result in a 170-dimensional vector.

$$f = (\zeta_{T=2} \cup \zeta_{T=3} \cup \dots \cup \zeta_Q) \quad (5)$$

It can also be observed from the equations that the length of the final vector is independent from the utterance length. Since the number of the frequency bins is fixed, the width of the rectangles will increase as the utterance length increases. Similarly, the width will decrease for utterances with short durations. The normalization in Equation 1 will minimize the duration effects.

Fig. 2 shows the extracted features from the respective spectrograms for randomly selected 2 spoofed utterances and 1 bonafide utterance. Log operation was applied to both the spectrograms and the energy features for a better illustration. Similarly, energy features vectors were reshaped into matrices for a better comparison with the spectrograms. Based on the value of T , it is possible to obtain a compact representation for the spectrogram, which is not possible with the traditional frame-based energies. These compact energy vectors can preserve the spoofing artifacts such as silent regions and can be combined with other features or deep embeddings to provide complementary information.

In this study, the extracted energy features are concatenated with the deep features obtained from the last layers of the deep models, and they are fed to the probability-to-similarity gradient (p2s-grad) based loss layer for the LCNN – LSTM model, which is a hyper-parameter-free discriminative loss function (Wang and Yamagishi, 2021), and fed to the softmax layer for the SE-Res2Net model. With this approach, all sub-band regional energy features are given to the network. After the training session, the network is expected to give more weight to the discriminative ones.

3. Spoof detection system

To investigate the performance of the regional energy features, different models are considered. Also, different feature parameters are used for different conditions. In this section, details of these systems are given.

3.1. LCNN – LSTM with LFCC features

3.1.1. LFCC

LFCC features are among the most popular features for audio spoof detection tasks. It is served as one of the baseline features for the recent ASVspoof 2021 challenge (Yamagishi et al., 2021). The parameters used in this work are given as follows. The speech signals are framed using a Hanning window of 20 ms length and 10 ms shift. Magnitude spectrograms are extracted using 1024 fast Fourier transform (FFT) bins. LFCC features are extracted from a linear filterbank with 20 filters. For the traditional LFCCs, the log energy feature is replaced with the zeroth feature. Delta and delta-delta features are appended. The final LFCC feature dimension is 60. Note that neither log energy nor zeroth feature is used with the proposed regional energy features. Although there is no restriction on using them together, the goal of this experiment is to prove that regional energy features are more discriminative than frame-based energies. Hence, the baseline system used LFCCs with log energies, and the other systems used LFCCs without log energies. All features are extracted using the full band (0 – 8 kHz).

Regional energy features are extracted as described in the previous section. Selected T values are {7, 11, 13, 15, 20, 25}. To examine the prime number combinations, $T=7$ and $T=11$, and $T=11$ and $T=13$ is considered. So, for the first combination, the energy feature vector length is 170 (49+121). For the second combination, the energy feature vector length is 290 (121+169).

3.1.2. LCNN – LSTM

LCNN is a very popular architecture for audio spoof detection (Lavrentyeva et al., 2017, Lavrentyeva et al., 2019, Wang and Yamagishi, 2021) and also served as one of the baselines for the recent ASVspoof 2021 challenge (Yamagishi et al., 2021). The LCNN used in this paper is based on the LCNN-LSTM-sum architecture given in Wang and Yamagishi (2021). The network includes the same LCNN from Lavrentyeva et al. (2019), but the layers after the maximum pooling layer is replaced with two Bi-LSTM layers and a skip connection is added over the Bi-LSTM layers. Mean pooling layer and a fully connected layer added as the last layers. The output of the fully connected layer is used as the embedding and fed to the p2s-gradient layer for classification. The embedding dimension is chosen as 64. The regional energy features are concatenated with the outputs of the mean pooling layer and fed to the fully connected layer which produces the embeddings for loss calculation. Therefore, the inclusion of the regional energy features only changes the number of inputs for the fully connected layer. All of the other parts of the system remain the same.

An important point worth mentioning is that the LCNN network receives varied-length input during training and inference, as in Wang and Yamagishi (2021). This approach eliminates the need for trimming or padding. The training and validation sets are ordered based on the utterance lengths. A batch size of 8 is used to group utterances with similar durations.

The p2s-gradient is a parameter-free solution to increase discriminative power of the network as given in Equation 6,

$$L = \frac{1}{|D|} \sum_{j=1}^{|D|} \sum_{k=1}^C (\cos \theta_{j,k} - 1(y_j = k))^2 \quad (6)$$

where $k \in \{1, \dots, C\}$ is the class label for the j -th trial over a dataset D . $\cos\theta$ value is computed as the cosine distance between the normalized weight vector for class k and the normalized utterance-level embedding vector. Equation 6 gives the mean-square error between the network's constrained output and the target scalar value.

3.2. SE-Res2Net with spectrogram features

3.2.1. Spectrogram

Although cepstral features are very popular among speech-related studies, with the modern deep learning models, spectrogram features are found to be effective. Many spoof detection systems used short-time Fourier transform based spectrograms or constant-Q transform based spectrograms (Lavrentyeva et al., 2019, Li et al., 2021, Lai et al., 2019, Xue and Zhou, 2023). In this study, two different approaches are followed to extract spectrogram features. The first is based on Zhang et al. (2021), where the spectrogram is extracted with 1728 window length, 130 hop length. Blackman window is used and only the lower half of the spectrogram is used, as it is reported that lower frequency region (0 – 4 kHz) is much more effective for synthetic speech detection. This configuration is used for LA conditions of ASVspoof 2019 and 2021 datasets, and also for DF condition of ASVspoof 2021 dataset. The final number of frequency bins is 433. It is important to note that the regional energy features are also extracted by using the lower half of the spectrogram.

For the PA condition of ASVspoof 2019, full frequency band is considered. The spectrogram is extracted with 400 window length, 160 hop length, and 512 FFT points as in Li et al. (2021). The extracted spectrograms have 257 bins, and the regional energy features are extracted from these spectrograms.

Contrary to the LFCCs, the spectrograms are extracted from utterances fixed to 6.4 seconds (shorter utterances are repeated, longer utterances are truncated) to follow the common practice (Zhang et al., 2021, Hua et al., 2021, Li et al., 2021, Xue and Zhou, 2023).

3.2.2. SE-Res2Net

Residual blocks (ResNet) (He et al., 2016) gave the researchers the opportunity to train very deep models. Res2Net (Gao et al., 2021) represented multi-scale features by constructing hierarchical residual-like connections. Squeeze-excitation (SE) blocks (Hu et al., 2018) assigns weights to channels to focus on the most informative ones. Using these mentioned blocks, many deep models were constructed and used for spoofed speech detection including SE-Res2Net50 (Li et al., 2021, Xue and Zhou, 2023), SE-DenseNet (Xue and Zhou, 2023), Res-TSSDNet (Hua et al., 2021), SENet34 (Lai et al., 2019), light-ResNet (Parasu et al., 2020). Due to its high performance, the SE-Res2Net50 model is used in this study. Two different concatenation strategies are utilized for regional energy features. The first one is directly concatenating the energy features with the output of the global average pooling layer of the SE-Res2Net, then applying a fully connected layer to reduce the dimension to two, followed by softmax layer. The second approach is applying a separate fully connected layer with ReLU activation to the energy features to reduce their dimension to 32, then concatenate its output with the output of the global average pooling layer. Table 1 shows the model architecture and regional energy concatenation paths. Direct concatenation is indicated by *dir*, concatenation after fully connected layer is indicated by *fc*. Fig. 3 visually summarizes the two approaches.

4. Experiments

4.1. Datasets

The proposed system's performance was tested using ASVspoof 2019 and ASVspoof 2021 datasets. The statistics of datasets are given in Table 2 below. ASVspoof 2019 LA set includes 19 text-to-speech and voice conversion attacks (A01 to A19). Six of them are included in the training set (A01 to A06), and others are included in the evaluation set. A16 and A19 are obtained with the same algorithms as A04 and A06, hence they are known attacks, while the remaining are the unknown attacks. PA set includes simulated replay attacks. Details of the simulation setup and other information can be found in Wang et al. (2020). For both partitions, bonafide – spoof classification is considered. For ASVspoof 2021 dataset, there is no training and development partitions, and participants are expected to use 2019 set for the training purposes. The 2021 dataset is intended to be more challenging than the previous datasets. For the LA task, transmission effects are introduced, for the DF task lossy codecs are used to encode and decode the audio, for the PA task recordings collected in various physical spaces are used (Yamagishi et al., 2021).

Table 1
SE-Res2Net50 (Li et al., 2021) architecture with regional energy features.

Stage	Direct Concatenation (Se-Res2Net50_dir)	Concatenation after FC layer (Se-Res2Net50_fc)
Conv1		[conv2d, 3×3, 16, stride=1] × 3
Conv2		[SE-Res2Net block, 16] × 3
Conv3		[SE-Res2Net block, 32] × 4
Conv4		[SE-Res2Net block, 64] × 6
Conv5		[SE-Res2Net block, 128] × 3
Average		Global Average Pool
Concat.	Concatenate with regional energies	32-d fc layer for regional energies
Output		Concatenate with 32-d fc outputs
		2-d fc layer, softmax

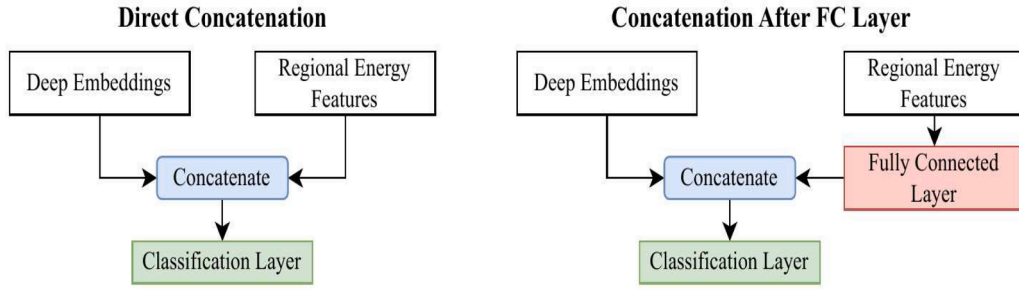


Fig. 3. Two approaches for concatenating regional energy features with deep embeddings. SE-Res2Net model was used in the experiments to obtain deep embeddings, hence the names SE-Res2Net_dir and SE-Res2Net_fc.

Table 2

Statistics of ASVspoof 2019 and ASVspoof 2021 datasets.

Partition	ASVspoof 2019						ASVspoof 2021	
	Train		Development		Evaluation		Evaluation	
	Bonafide	Spoof	Bonafide	Spoof	Bonafide	Spoof	Bonafide	Spoof
LA	2580	22800	2548	22296	7355	63882	14816	133360
PA	5400	48600	5400	24300	18090	116640	94068	627264
DF	-	-	-	-	-	-	14869	519059

4.2. Training parameters

PyTorch toolkit was used for implementing all deep learning related operations. For the LCNN – LSTM models, Adam optimizer with default parameters was used. The initial learning rate was chosen as $3e-4$ and halved after every ten epochs. Batch size was 8 and each batch consists of similar duration signals as described before. The p2s-gradient loss function was used. The models were trained for 150 epochs with an early stop option if the EER for the development set does not decrease for 15 epochs. The models with the lowest EER for the development set were used in the evaluation stage. The initial weights have a high impact on the results. To overcome this issue, a common practice is to repeat the experiment with a different random seed (Wang and Yamagishi, 2021, Jung et al., 2022). To this end, each model was trained three times with the random seed for the k^{th} run set to 10^{k-1} as in Wang and Yamagishi (2021).

For the SE-Res2Net models, the same procedure of Li et al. (2021) was followed, using Adam optimizer with warm-up steps. Weighted cross-entropy loss was used to train the models for 20 epochs, and models with the lowest EER on the development set are used in the evaluation. Random seed was set to 0 in this case.

4.3. Experimental results for LCNN – LSTM

The first set of experiments were focused on determining the best T values for the regional energy features. LFCC features and LCNN – LSTM models were used for this purpose on the ASVspoof 2019 dataset. Table 3 shows the results for different T values, in terms of EER (%). EER can be defined as the point where the false acceptance rate is equal to the false rejection rate. The proposed approach is more effective than the traditional frame-based energy features for the evaluation partitions of both LA and PA sets. On the other hand, the difference between the best and the average EER values for different random seeds emphasizes the importance of the initial weights, as there was no other source of randomness.

The best performance for the LA set was obtained with $T=13$. A 20% relative EER reduction was achieved compared to the LFCC

Table 3

Comparison of LCNN – LSTM models in terms of EER (%). Reported values are the best performing seeds, values inside parentheses are the average of the different random seeds.

Features	LA	PA
LFCC with log energy	2.019 (2.592)	2.067 (2.154)
LFCC + regional energy ($T=7$)	1.904 (2.209)	1.841 (2.268)
LFCC + regional energy ($T=11$)	2.232 (2.528)	1.664 (1.782)
LFCC + regional energy ($T=13$)	1.619 (2.015)	2.139 (2.206)
LFCC + regional energy ($T=15$)	1.845 (2.906)	1.576 (1.771)
LFCC + regional energy ($T=20$)	1.926 (2.736)	1.443 (1.911)
LFCC + regional energy ($T=25$)	2.039 (2.979)	1.509 (1.874)
LFCC + regional energy ($T=7 + T=11$)	1.795 (2.385)	1.874 (2.106)
LFCC + regional energy ($T=11 + T=13$)	2.260 (2.661)	1.631 (1.883)

with log energy features in the best case. For the average of the random seeds, 22% relative EER reduction was observed. An interesting observation is that four of the proposed features performed better than the baseline, and the other four performed under the baseline, comparing the average EERs. However, comparing the best case EERs, only two of the proposed features performed worse than the baseline. This situation also stresses the importance of initial model parameters.

For the PA partition, the best EER result was obtained with $T=20$. This corresponds to a relative EER reduction of about 30%. The best average EER was obtained with $T=15$, which leads to an 18% relative EER reduction. The comments made about the initial values are also true for the PA set. In this case, however, only $T=13$ underperformed in terms of best EER, and only two of the proposed approaches performed worse than the baseline in terms of average EER.

The results showed that concatenating the features from different prime numbers did not bring much advantage (last two rows of Table 2), contrary to the texture example given in Guido (2016). Instead, selecting a suitable T value seems to be more important since the best performing value for the LA set became the worst performing one for the PA set. Finally, the number of learnable parameters was 275,968 for the LCNN – LSTM model, which increased to 289,784 for $T=13$ and to 301,568 for $T=20$.

4.4. Experimental results for SE-Res2Net

Based on the results of the initial experiments, $T=13$ was chosen for LA and DF tasks, and $T=15$ was chosen for PA tasks in further experiments using SE-Res2Net models. As a reminder, for the LA and DF tasks a total of 433 bins were used (lower half after the 1728-point FFT) while PA task 257 bins were used (full band after 512-point FFT). The baseline SE-Res2Net model, which did not incorporate regional energy features, was also trained from scratch and the observed results were different from Li et al. (2021), probably due to the hardware/software differences.

Table 4 shows the results for the ASVspoof 2019 LA and PA sets. For both tasks, SE-Res2Net_fc achieved the best evaluation EER. Compared to the SE-Res2Net model, 6.3% relative EER reduction was achieved for the PA task. A 7.2% relative EER reduction was achieved for the LA task. The computational load, on the other hand, was negligible. SE-Res2Net had 923,102 learnable parameters, while SE-Res2Net_fc had 930,398 learnable parameters for PA task ($T=15$) and had 928,606 parameters for LA task ($T=13$). The learnable parameters only increased 0.79% and 0.59% with the proposed approach for PA and LA tasks, respectively.

Direct concatenation performed worse than the baseline. For the LCNN – LSTM model, the concatenated vector was given to the fully connected layer to extract embedding for p2s-gradient loss calculation. SE-Res2Net_dir sends the concatenated vector to the softmax layer after reducing the dimension to two. The deep embeddings after the global average pooling layer were 128-dimensional, while the regional energy features were 169- and 225-dimensional (for LA and PA tasks, respectively). The regional energy features were intended to be complementary features and they do not have the same discriminative power as the deep embeddings. So, directly concatenating more energy features than the deep embedding features for classification is not feasible, as verified experimentally. However, even a single fully connected layer such as in LCNN – LSTM or SE-Res2Net_fc models is sufficient to exploit the regional energy features.

4.5. Comparison with other systems

The proposed systems were compared with the other studies in this subsection. For the PA task, although the proposed features increased the performance compared to the baseline, their performances fell behind the state-of-the-art. For the LA task, considerably better performances were obtained. Hence, a more detailed analysis for LA task is presented, providing comparisons with the state-of-the-art models for both pooled and attack based EERs.

The results of the proposed models and other single systems from the literature are given in Table 5 for the PA task, and in Table 6 for the LA task. The results are sorted by the min t-DCF values, which was the primary performance metric for the challenge (Kinnunen et al., 2020). It assesses the impacts of CM on automatic speaker verification performance, where the necessary parameters related to the speaker verification are given by the organizers. The EER values are also given as the secondary metric.

As seen in Table 6, LCNN – LSTM model with the proposed energy features achieved the best score for the LA task among the other LCNN based systems. SE-Res2Net_fc where the regional energies concatenated to the deep embeddings after a fully connected layer achieved one of the best performances in terms of both t-DCF and EER. Direct concatenation (SE-Res2Net_dir) performed slightly worse than the same model without the regional energy features. The proposed approach also performed better than the most recent dynamic modulated Res2Net (DM-Res2Net) (Yang et al., 2023) model which also used low frequency spectrogram features.

EERs for each attack are presented in Table 7 for the SE-Res2Net models and state-of-the-art models. None of the models consistently outperformed the others. DFSincNet and SE-Res2Net_fc achieved the lowest EERs for 4 different attack types out of 13. SE-Res2Net models could not accurately detect A08 attacks, compared to the other models. SE-Res2Net_fc model also performed bad for

Table 4
EER (%) results for SE-Res2Net models.

Model	LA		PA	
	Dev	Eval	Dev	Eval
SE-Res2Net	0.15	0.97	1.59	2.06
SE-Res2Net_dir	0.08	1.04	1.97	2.49
SE-Res2Net_fc	0.16	0.90	1.74	1.93

Table 5

Comparison of the proposed system with others for the PA set in terms of min t-DCF and EER (%).

Model	Min t-DCF	EER
SE-Res2Net Xue and Zhou (2023)	0.023	0.85
CQTMGD – ResNeWt Cheng et al. (2019)	0.025	0.94
CQT – LCNN Lavrentyeva et al. (2019)	0.03	1.23
LCNN – LSTM ($T=20$)	0.044	1.44
SE-Res2Net_fc ($T=15$)	0.045	1.93
MGD – ResNeWt Cheng et al. (2019)	0.047	2.15
SE-Res2Net	0.051	2.06
ATCoP+GTCC+SVM Javed et al. (2022)	0.064	1.00
SE-Res2Net_dir ($T=15$)	0.064	2.49
LCNN Wei et al. (2022)	0.074	2.47
LFCC – LCNN Lavrentyeva et al. (2019)	0.105	4.60

Table 6

Comparison of the proposed systems with others for the LA set, in terms of min t-DCF and EER (%).

Model	# Parameters (K)	Min t-DCF	EER
DFSincNet Huang et al. (2023)	433.7	0.017	0.52
SE-Res2Net_fc ($T=13$)	928.6	0.026	0.90
SE-Res2Net	923.1	0.027	0.97
AASIST Jung et al. (2022)	297	0.028	0.83
Rawnet2 Hansen and Wang (2022)	-	0.028	0.99
SE-Res2Net_dir ($T=13$)	923.4	0.030	1.04
RawGAT-ST Tak et al. (2021)	437	0.034	1.06
DM-Res2Net Yang et al. (2023)	950	0.036	1.21
SENet Zhang et al. (2021)	1,100	0.037	1.14
L-DenseNet Li et al. (2022)	338.3	0.045	1.54
LCNN – LSTM ($T=13$)	286.7	0.048	1.62
Res-TSSDNet Hua et al. (2021)	350	0.048	1.64
LCNN – LSTM Wang and Yamagishi (2021)	275.9	0.052	1.92
SE-DenseNet Xue and Zhou (2023)	-	0.074	2.82
LCNN – Dual attention Ma et al. (2021)	-	0.078	2.76
LFCC – LCNN Lavrentyeva et al. (2019)	-	0.100	5.06

A11 and A17 attack types. Other than these attacks, the proposed model performed similar or even better than the others.

4.6. Results for ASVspoof 2021 dataset

The ASVspoof 2021 challenge dataset contains mismatched data, as the training and validation process are realized using ASVspoof 2019 data, and evaluation data for the LA and DF tasks contains transmission and compression effects. To achieve a high performance, data augmentation is necessary, which reduces the mismatch between the training and test data ([Tomilov et al., 2021](#), [Huang et al., 2023](#), [Cáceres et al., 2021](#)). Data augmentation methods such as frequency masking, creating additional training data, using mixup techniques, etc. are among the preferred methods. In this study, data augmentation is not performed as the regional energy features may need some modifications to suit the chosen augmentation method. Nevertheless, results with no augmentation are presented to verify that the additional regional energy features can still perform better than the baseline system. Also, results from other systems with no data augmentation are also given for comparison.

[Table 8](#) shows the results for the ASVspoof 2021 LA set. It is beneficial to note that the proposed models were the same models used in the ASVspoof 2019 LA evaluation (same models were also used for the DF task given below). Without any data augmentation, the proposed models and others performed worse than the baseline models provided by the organizers (LFCC – LCNN and RawNet2 models). Among the proposed models, SE-Res2Net_fc achieved the lowest EER, but SE-Res2Net model achieved the lowest t-DCF value. The direct concatenation strategy again performed worse than the others.

Table 7

Attack based EERs (%) for SE-Res2Net models and state-of-the-art systems (ASVspoof 2019 LA set).

Model	A07	A08	A09	A10	A11	A12	A13	A14	A15	A16	A17	A18	A19	Pooled
SE-Res2Net	0.14	4.54	0.16	0.47	0.50	0.28	0.24	0.12	0.30	0.47	1.1	0.96	0.51	0.97
SE-Res2Net_dir	0.38	1.71	5.17	0.53	0.65	0.30	0.12	0.12	0.49	0.46	1.30	0.99	0.46	1.04
SE-Res2Net_fc	0.14	1.56	0.06	0.46	1.44	0.10	0.06	0.09	0.36	0.38	1.83	0.85	0.72	0.90
RawNet2 Hansen and Wang (2022)	0.91	0.29	0.02	1.39	0.31	1.30	0.08	0.18	0.61	0.24	2.11	2.25	0.97	0.99
AASIST Jung et al. (2022)	0.8	0.44	0.0	1.06	0.31	0.91	0.1	0.14	0.65	0.72	1.52	3.40	0.62	0.83
DFSincNet Huang et al. (2023)	0.37	0.08	0.04	0.76	0.49	0.65	0.04	0.18	0.39	0.29	1.02	0.63	0.65	0.52

Table 9 shows the results for the ASVspoof 2021 DF set, without any data augmentation. For this task, the performance metric was EER, as it does not include an automatic speaker verification system (Yamagishi et al., 2021). Although there are not many studies reporting DF performance without data augmentation, Table 8 shows that the proposed method cannot surpass baseline models, although the difference is modest. Interestingly, both concatenation strategies outperformed the SE-Res2Net. For the SE-Res2Net_fc model, relative EER reduction was 20.73%.

4.7. Combination with other features

The experiments given above showed that proposed energy features can be combined with LFCC and log power spectrogram features (to be more accurate, combined with deep embeddings extracted from these features). There are many other feature types besides them. In order to verify that the proposed approach is not limited to LFCC and STFT based spectrogram, further experiments were conducted on ASVspoof 2019 set using MFCCs, log mel-filterbank energies, and CQCCs. SE-Res2Net_fc configuration was used as it generally gave better performance.

For the LA condition, 60-dimensional features were used for MFCC and CQCC (including 20 static features and their delta and acceleration coefficients). 50 filters were used for mel-filterbank energies. Only the lower half band was considered for MFCC and mel-filterbank features, while the full band was considered for the CQCCs. T was chosen as 13.

For the PA condition, the same MFCC and mel-filterbank configurations were used as in the LA case but the full band was used. On the other hand, only 30 static features were used for the CQCCs. T was chosen as 15. Other training parameters were the same as given in the previous sub-sections. The results were given in Table 10 below.

From the results, it is clear that the LA condition had benefited from the energy features more than the PA condition. This may be due to the fact that the speech synthesis methods cannot accurately synthesize “silence”. The spoofing artifacts from the synthesized speech may therefore be preserved in the regional energy features. The artifacts for the PA condition may not be as obvious as for the LA condition. Yet using the energy features improved the performance in some cases. These observations also verify the previous experiments realized with SE-Res2Net model. Although there were some improvements, chosen models and features could not deliver impressive results for the PA condition. On the other hand, the proposed approach could compete with the state-of-the-art for the LA condition, outperforming many other systems from literature. Nevertheless, the relative improvement for the PA condition with LCNN – LSTM model was higher than the relative improvement for the LA condition. Given the negligible cost of the proposed approach, it may be worth utilizing for a potential performance boost.

5. Discussion

The results presented in the previous section showed that the regional energy features can work with both different feature types, different deep learning models, and different spectrogram configurations. The assumption that spoofing artifacts can be captured with complementary regional energy features was validated experimentally. In this section, observed results are discussed and possible future studies are addressed.

LCNN – LSTM model with regional energy features have achieved a high performance for both LA and PA conditions. For the best cases, 20% and 30% relative EER reductions for the LA and PA tasks were achieved with a negligible increase in the model size. Different vector sizes for energy features led to different performances. Using a small dimensional vector or a large dimensional vector seems to deteriorate the performance. T values between 13 and 20 are found to be a sweet spot, empirically.

Based on the observations via LCNN – LSTM model, SE-Res2Net models were used in further experiments with spectrogram features. In this case, different FFT points were used for LA and PA tasks, following the reported results in the literature. As lower frequency was found to be more effective for LA tasks, 433 frequency bins were used as described previously. For $T=13$, this number leads to a frequency span of 33 Hz per rectangle. For the LCNN – LSTM model, the span was 39 Hz ($T=13$) as 512 bins were used in that configuration. For the PA task with SE-Res2Net model, 257 bins were used, which corresponds to 17 Hz for $T=15$. For the LCNN – LSTM model, the span was 25 Hz, as 512 bins with $T=20$ was used in the best case. These differences show that the regional energy features can work under various common FFT configurations.

The relative changes in EER and parameter sizes are given in Table 11 for a better summarization. For the LCNN – LSTM model, as high relative EER reductions as 20% and 30% were achieved with only 5% and less than 10% increment in the model size, for LA and PA tasks, respectively. For SE-Res2Net_fc model, more than 5% improvements were observed for less than 1% model size increment. This situation verifies that the complementary features are highly cost effective.

For the ASVspoof 2021 LA and DF tasks, SE-Res2Net_fc model achieved lower EERs than the SE-Res2Net, but the need for data augmentation is clear as the obtained results could not even pass the challenge baselines. This situation indicates that the spoof artifacts discussed in the Introduction section vanish when transmission and codec effects are present. Data augmentation introduces the same effects to the training data, thus regional energy features may be able to capture useful information again. Future studies will focus on how to use regional energies when data augmentation methods are present.

As a final note, the proposed energy features were also used with the AASIST model (Jung et al., 2022), but slightly worse performances were observed and the results were not presented here. A possible reason for this is the fact that AASIST model consists of different parts, including graph attention networks applied to spectral and temporal dimensions separately. Graph attention layers act as a data-driven attention mechanism, extracting the most discriminative information. The regional energy features may not further increase the discriminative power of the model. Also, both AASIST and DFSincNet models use SincConv as the first layers to obtain 2-D spectro-temporal feature maps. Extracting features similar to the regional energies from those feature maps may also be useful, which

Table 8

Results for ASVspoof 2021 LA set without data augmentation, in terms of min t-DCF and EER (%).

Model	Min t-DCF	EER
LFCC – LCNN Yamagishi et al. (2021)	0.3152	8.9
RawNet2 Yamagishi et al. (2021)	0.4152	9.49
SE-Res2Net	0.4458	11.20
SE-Res2Net_fc ($T=13$)	0.4645	10.99
SE-Res2Net_dir ($T=13$)	0.5151	12.91
LFCC – ASSERT34 Benhafid et al. (2021)	0.603	16.98
Light-TDNN-CE Cáceres et al. (2021)	0.6129	17.89

Table 9

Results for ASVspoof 2021 DF set without data augmentation, in terms of min t-DCF and EER (%).

Model	EER
MSTFT – LCNN Tomilov et al. (2021)	21.9
RawNet2 Yamagishi et al. (2021)	22.38
LFCC – LCNN Yamagishi et al. (2021)	23.48
SE-Res2Net_fc ($T=13$)	24.39
SE-Res2Net_dir ($T=13$)	27.21
SE-Res2Net	30.77

Table 10

Results for ASVspoof 2019 set with different feature types EER (%).

Model	Feature	LA		PA	
		Dev	Eval	Dev	Eval
SE-Res2Net	Log power magnitude	0.15	0.97	1.59	2.06
	Log mel-filterbank	0.71	4.30	2.71	3.60
	MFCC	0.75	10.54	6.28	9.20
	CQCC	0.04	13.48	5.02	7.69
SE-Res2Net_fc	Log power magnitude	0.16	0.90	1.74	1.93
	Log mel-filterbank	0.59	3.14	2.50	4.03
	MFCC	0.75	9.92	6.30	10.60
	CQCC	0.04	13.57	5.29	7.20

Table 11

Relative changes in EER and parameter sizes for the proposed models compared to LCNN – LSTM and SE-Res2Net baseline models (ASVspoof 2019 evaluation set).

Model	Relative reduction in EER		Relative increase in model size	
	LA	PA	LA	PA
LCNN – LSTM	19.81%	30.19%	5.00%	9.27%
SE-Res2Net_fc	7.21%	6.31%	0.59%	0.79%
SE-Res2Net_dir	-7.21%	-20.87%	0.03%	0.05%

will be investigated in future studies.

6. Conclusions

This study applied regional energy features to spoofed speech detection tasks. Several different configurations were considered based on the selected LA/PA tasks and deep learning models. ASVspoof 2019 dataset was used for the experiments, which showed that the proposed models can achieve comparable results to the state-of-the-art with SE-Res2Net configuration for the LA task. Experiments with the ASVspoof 2021 LA and DF sets without any data augmentation indicated that the proposed models could improve the results compared to the same models without the regional energy features. However, data augmentation methods are necessary to achieve a competitive performance. For LCNN – LSTM models, relative EER reductions around 20% and 30% were observed with the cost of 5% and 10% increased model sizes for LA and PA tasks of 2019 challenge, respectively. For SE-Res2Net models, more than 7% and 6% relative EER reductions were achieved with less than 1% increased model sizes for LA and PA tasks, respectively. Therefore, the proposed features are low cost and can be used complementary to spectral or cepstral features. Future studies will focus on extracting regional features with data augmentation methods to achieve a higher performance under the more challenging ASVspoof 2021 data

and extracting similar features by using the feature maps obtained via SincConv layers instead of spectrograms.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data is publicly available.

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